

Malware Analysis



CYBERTOOLS



What is Malware Analysis?

Malware analysis is the process of examining malicious software to understand its behavior, functionality, and impact. This analysis involves dissecting the malware to determine how it operates, what it targets, and how it spreads. The primary goal is to identify the malware's purpose, origins, and the best methods to defend against it.



What Are Malware Analysis Tools?

Malware analysis tools are specialized software and platforms designed to facilitate the examination and breakdown of malware. These tools can be categorized into several types:

1	Static Analysis Tools	Analyze the malware without executing it, inspecting its code and structure.
2	Dynamic Analysis Tools	Observe the malware in action by running it in a controlled environment to monitor its behavior.
3	Hybrid Analysis Tools	Combine static and dynamic analysis techniques for a comprehensive evaluation.
4	Memory Analysis Tools	Examine the system's memory to detect and analyze malware that may be running in RAM.

Why Do We Need Malware Analysis Tools?

Malware analysis tools are essential for several reasons:

1. **Threat Identification:** Understanding new and evolving malware threats allows security professionals to develop effective defenses.
2. **Incident Response:** Quickly analyzing malware involved in security incidents helps mitigate damage and recover systems.
3. **Vulnerability Detection:** Identifying how malware exploits system vulnerabilities aids in patching and securing those weaknesses.
4. **Improving Defenses:** Continuous analysis and understanding of malware inform the development of better security protocols and software updates.
5. **Forensic Investigations:** Providing critical insights during cybercrime investigations to understand how attacks were carried out.

8 Tools For Malware Analysis



1. Cuckoo Sandbox

Cuckoo Sandbox is an advanced, highly modular, and 100% open-source automated malware analysis system.

Tool: Cuckoo Sandbox



2. YARA

YARA, standing for “Yet Another Recursive Acronym,” is an open-source malware analysis tool used to analyze individual malware samples based on textual or binary patterns.

Tool: YARA



3. VMRay Analyzer

VMRay Analyzer is a malware analysis platform that uses intelligent monitoring to distinguish between malicious behavior and legitimate activity.

Tool: VMRay Analyzer



4. VIPRE Threat Analyzer Sandbox

VIPRE Threat Analyzer Sandbox is a dynamic malware analysis sandbox that allows safe examination of the potential impact of malware on an organization.

Tool: VIPRE Threat Analyzer Sandbox



5. GHIDRA

GHIDRA is an open-source software reverse engineering framework developed by the National Security Agency (NSA).

Tool: GHIDRA



6. CrowdStrike Falcon Sandbox

CrowdStrike Falcon Sandbox is a malware analysis solution that enables enterprises to conduct in-depth analysis of unknown and zero-day threats.

Tool: CrowdStrike Falcon Sandbox



7. IDA Pro

IDA Pro is a powerful disassembler and versatile debugger, capable of creating execution maps to display the binary instructions processed by the CPU symbolically.

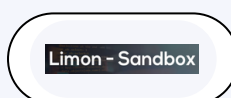
Tool: IDA Pro



8. Limon — Sandbox

Limon is a sandbox developed as a research project written in Python, which automatically collects, analyzes, and reports on the runtime indicators of Linux malware.

Tool: Limon — Sandbox



8 Books About Malware



CYBER BOOKS

1. Malware Analysis and Detection Engineering

Authors

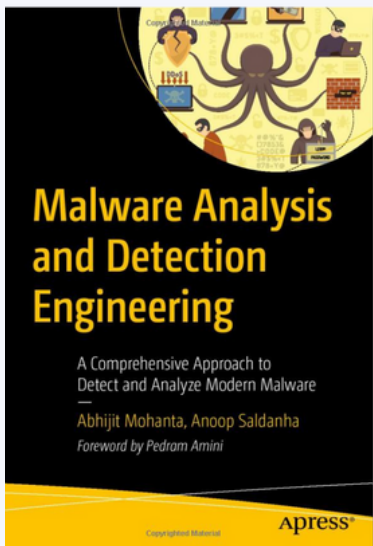
Abhijit Mohanta, Anoop Saldanha

Description

Malware Analysis and Detection Engineering is a one-stop guide to malware analysis that simplifies the topic by teaching you undocumented tricks used by analysts in the industry.

What You Will Learn

- Analyze, dissect, reverse engineer, and classify malware
- Effectively handle malware with custom packers and compilers and more.



[Get Book >](#)

2. Mastering Malware Analysis

Authors

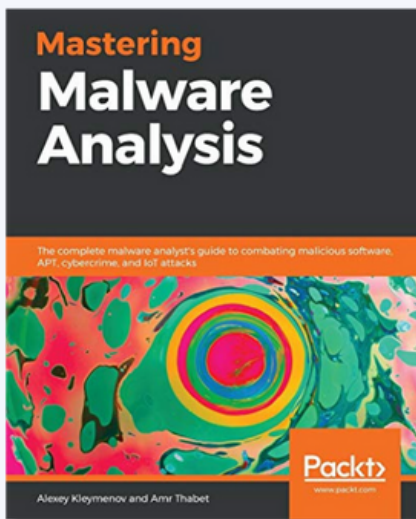
Alexey Kleymentov, Amr Thabet

Description

Mastering Malware Analysis explains the universal patterns behind different malicious software types and how to analyze them using a variety of approaches.

What You Will Learn

- Explore widely used assembly languages to strengthen your reverse-engineering skills
- Master different executable file formats, programming languages, and relevant APIs used by attackers and more.



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3. The Art Of Memory Forensics

Authors

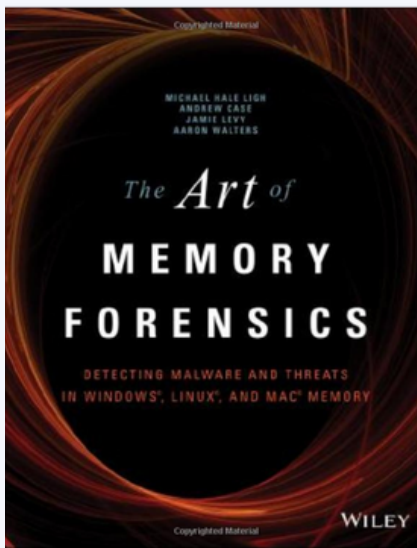
Michael Ligh, Andrew Case, Jamie Levy, Aaron Walters

Description

The Art of Memory Forensics is based on a five day training course that the authors have presented to hundreds of students. It focuses exclusively on memory forensics and how to deploy such techniques.

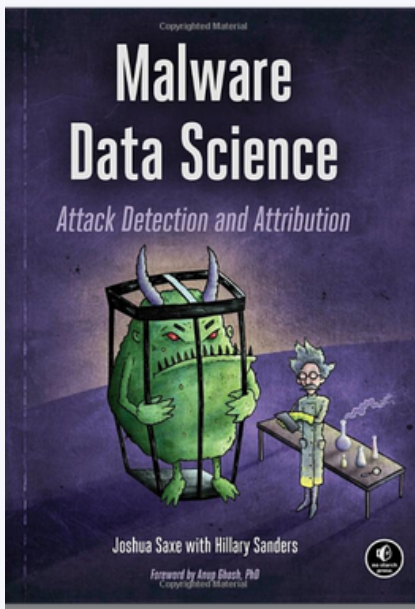
What You Will Learn

- How volatile memory analysis improves digital investigations
- Proper investigative steps for detecting stealth malware and advanced threats and more.



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4. Malware Data Science



Authors

Joshua Saxe, Hillary Sanders

Description

Malware Data Science introduces machine learning, statistics, social network analysis, and data visualization, and shows how to apply these methods to malware detection and analysis.

What You Will Learn

- Analyze malware using static analysis
- Observe malware behavior using dynamic analysis
- Identify adversary groups through shared code analysis and more.

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5. Rootkits and Bootkits

Authors

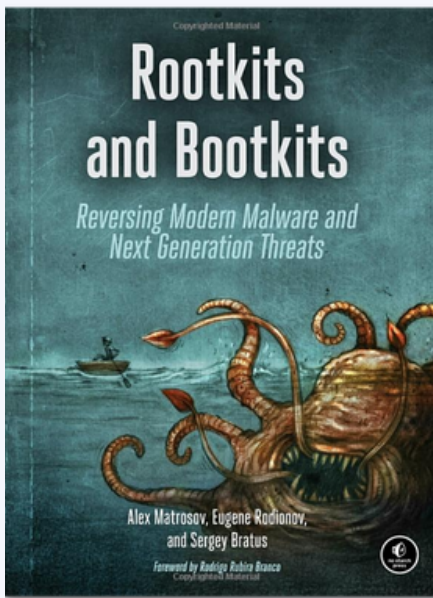
Alex Matrosov, Eugene Rodionov,
Sergey Bratus

Description

Rootkits and Bootkits will teach you how to understand and counter sophisticated, advanced threats buried deep in a machine's boot process or UEFI firmware.

What You Will Learn

- How Windows boots and UEFI mode—and where to find vulnerabilities
- The details of boot process security mechanisms like Secure Boot and more.



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6. Learning Malware Analysis

Authors

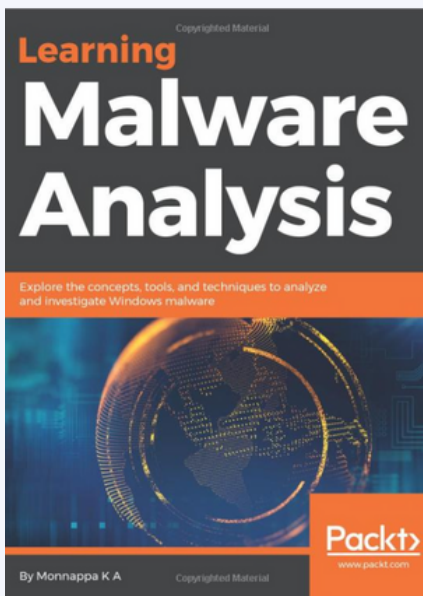
Monnappa K A

Description

This book teaches you the concepts, techniques, and tools to understand the behavior and characteristics of malware through malware analysis. It also teaches you techniques to investigate and hunt malware using memory forensics

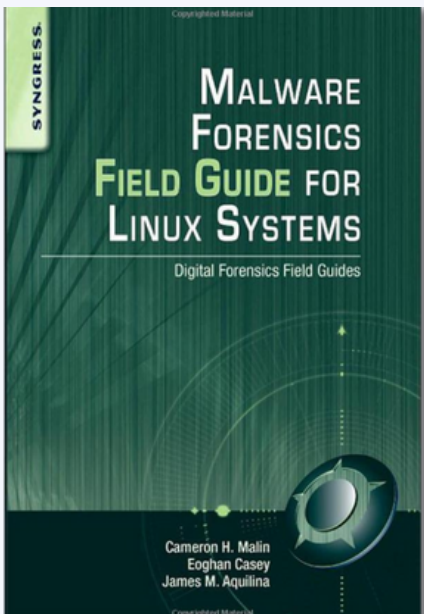
What You Will Learn

- Concepts, techniques, and tools to understand the behavior and characteristics of malware
- Techniques to investigate and hunt malware using memory forensics and more.



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7. Malware Forensics Field Guide For Linux Systems



Authors

Cameron H. Malin, Eoghan Casey,
James M. Aquilina

Description

Malware Forensics Field Guide for Linux Systems is a handy reference that shows students the essential tools needed to do computer forensics analysis at the crime scene.

What You Will Learn

- Malware incident response – volatile data collection and examination on a live Linux system
- Analysis of physical and process memory dumps for malware artifacts and more.

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8. Hacking Malware and Rootkits Exposed

Authors

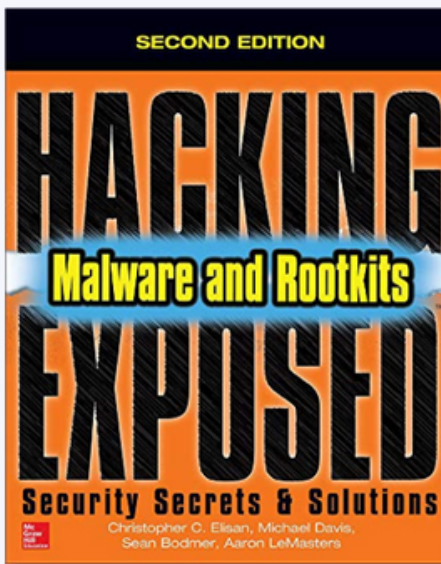
by Christopher Elisan, Michael Davis, Sean Bodmer & Aaron LeMasters

Description

This book fully explains the hacker's latest methods alongside ready-to-deploy countermeasures. Discover how to block pop-up and phishing exploits, terminate embedded code, and identify and eliminate rootkits.

What You Will Learn

- Learn how malware infects, survives, and propagates across an enterprise
- See how hackers develop malicious code and target vulnerable systems and more.



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Each of these tools and books offers unique capabilities and features, making them invaluable assets in the fight against cyber threats. By leveraging these tools, cybersecurity professionals can effectively analyze and respond to malware, ensuring robust protection for their systems and data.

Want more insights like this?



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